

Dr. Qusay Abeed, PhD

Geological Director



GDAC-SA Tender Role: GDAC Technical Director | Native Arabic Speaker | Halliburton Petroleum Systems Lead | CCUS Expertise (Vision 2030 Alignment)

GDAC Technical Director - Petroleum Geochemist

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Current Positions

Petroleum Geochemist Senior / Permedia and DSG Petroleum Systems Lead QA

Halliburton | United Kingdom *March 2020 - Present (5 years)*

- Senior role overseeing petroleum systems quality assurance for Permedia and DSG products
- Expertise in C++ programming and Carbon Capture, Utilisation and Storage (CCUS)
- Technical leadership for petroleum systems modeling and interpretation

Permedia CO2 Team Agile Scrum Master

Landmark Software (Halliburton) | Oxford, England, United Kingdom (Hybrid) *March 2017 - Present (8 years)*

- Dealing with & developing technologies for basin modelling and CO2 storage
- Leading Agile/Scrum practices for CO2 technology development team
- Technical expertise in C++ and Carbon Capture, Utilisation and Storage (CCUS)

Core Expertise

- Carbon Capture & Storage (CCUS)
- Petroleum Geochemistry
- Basin Modelling
- Petroleum Systems Analysis
- Source Rock Characterization
- Oil-Source Rock Correlation
- C++ Programming
- Agile/Scrum Project Management

Languages

- **Arabic:** Native speaker - Direct communication with Saudi Geological Survey stakeholders
- **English:** Fluent - International professional communication

Key Professional Experience

Petroleum Geochemist / Petroleum Systems Lead

Exploration Insights, Landmark, Halliburton | Abingdon, UK *October 2013 - March 2020 (6 yrs 6 mos)*

- Working as Petroleum System Lead, dealing with search, gathering, and processing petroleum systems data
- Global coverage: South America, Central America, North Africa, Lower 48 (USA) and Middle East
- **Created Global Petroleum Systems Interpretations Maps** for diverse regions
- Deep understanding of Middle East petroleum systems including Mesopotamian Basin
- Skills: Log Interpretation, Research and Development (R&D)

Petroleum Geologist / Post-Doctoral Researcher

RWTH Aachen University | Aachen, Germany *October 2012 - October 2013 (1 yr 1 mo)*

- Post-doctoral research on source rock/oil characterization, oil-source rock correlation, and oil-oil correlation
- Focus on bulk properties, biomarkers and molecular geochemistry of oil
- **Basin modelling for the Mesopotamian Basin of Iraq** - one of the largest petroleum systems in the world
- Gained experience in managing petroleum system modelling and seismic interpretation

Research Associate

RWTH Aachen University | Aachen, Germany *October 2008 - October 2012 (4 yrs 1 mo)*

- Petroleum systems research at Institute of Geology and Geochemistry of Petroleum and Coal
- Advanced research in organic geochemistry and thermal/burial history
- Skills: Research and Development (R&D), Presentations

Researcher and Exploration Instructor

Tikrit University | Tikrit, Iraq *November 2006 - May 2008 (1 yr 7 mos)*

- Researcher at the Institute of Geology and Geochemistry of Petroleum and Coal at RWTH Aachen University
- Main field of interest: Organic geochemistry of petroleum and source rocks
- Thermal/burial history and basin modelling research
- Skills: Research and Development (R&D), Presentations

Education

PhD - Petroleum Geochemistry & Basin Modelling

RWTH Aachen University | Aachen, Germany *2008 - 2012*

- **Grade: Magna Cum Laude**
- Thesis focused on Mesopotamian Basin petroleum systems

MSc - Petroleum Geology

University of Baghdad | Baghdad, Iraq *2002 - 2006*

- **Grade: 81.91%**

BSc - Geology/Earth Science

University of Baghdad | Baghdad, Iraq *1998 - 2002*

- **Grade: 71.7%**

Technical Skills

- **CCUS Technologies:** Carbon capture, utilisation and storage systems - aligns with Saudi Vision 2030 sustainability initiatives
- **Programming:** C++ for petroleum systems modeling and basin simulation
- **Geochemical Analysis:** Biomarkers, molecular geochemistry, source rock characterization
- **Basin Modelling:** Thermal history, burial history, petroleum systems modeling
- **Seismic Interpretation:** 2D/3D seismic data interpretation
- **Log Interpretation:** Well log analysis and correlation
- **Project Management:** Agile/Scrum methodologies

Geographic Expertise

- **Middle East:** Mesopotamian Basin (Iraq), North Africa - deep regional knowledge
- **Americas:** South America, Central America, Lower 48 (USA)
- **Europe:** UK, Germany - extensive work experience

Relevance to GDAC-SA Tender

- **Native Arabic Speaker:** Direct, fluent communication with Saudi Geological Survey stakeholders and government officials
- **Middle East Geology Expertise:** Deep understanding of regional petroleum systems including Mesopotamian Basin research
- **Halliburton Connection:** Former colleague of Dr. Fabian Kohlmann with shared company culture and work methodology
- **CCUS Knowledge:** Expertise in carbon capture and storage aligns perfectly with Saudi Arabia’s sustainability goals and Vision 2030 initiatives
- **Global Petroleum Systems Experience:** Created interpretations for diverse regions including Middle East, North Africa, and Americas
- **Basin Modelling Expertise:** Essential technical capability for geological data platform development and analysis
- **Programming Skills:** C++ expertise valuable for database development and system integration
- **Academic Excellence:** PhD Magna Cum Laude from prestigious RWTH Aachen University
- **Project Management:** Agile Scrum Master experience for effective team coordination
- **Research Background:** Strong R&D capabilities from academic and industry research roles